Open Problems and System Cases

Open questions connect model capabilities, reusable execution, temporal evidence, and task outcomes.

Open problems in agent construction

Each question connects a system capability to the evidence needed to evaluate it.

Temporal perception and execution	Connect temporal representations to precise numerical operations and checked outputs.	LLM + GTS
Memory under changing information	Track availability, revisions, and changing regimes when reusing past experience.	GH + GTS
Transfer of learned procedures	Test reusable procedures across datasets, domains, and task contracts.	GH + GTS
Evidence-sensitive evaluation	Measure how information quality, tool use, and validation affect the final result.	GH + GTS
Prediction-to-decision feedback	Evaluate actions, delayed outcomes, uncertainty, and the cost of prediction errors.	TSK

Focused system cases

Examples connect capabilities to their use in a system.

ChatTS — LLM side

Temporal alignment and dialogue supervision make numerical series available to language reasoning.

TimeClaw — General Harness

Executed trajectories are compared and distilled into reusable experience.

ModernTSF — Infrastructure

Forecasting models and datasets provide resources for task evaluation.

Source passages connect each case to its claims, modules, and task requirements.

Evidence record

Claim | **Module** | **Task contract** | **Evidence locator**